Simulated impact of a fish based shift in the population n--3 fatty acids intake on exposure to dioxins and dioxin-like compounds.

Due to the favourable health effects of LC n-3 PUFAs, marine products have been recognised as a food group of special importance in the human diet. However, seafood is susceptible to contamination by lipophilic organic pollutants. The objective of this study was to evaluate intake levels of PCDDs, PCDFs and dioxin-like PCBs, by a probabilistic Monte Carlo procedure, in relation to the recommendation on LC n-3 PUFAs given by Belgian Federal Health Council. Regarding the recommendation, two scenarios were developed differing in LC n-3 PUFAs intake: a 0.3 E% and a 0.46 E% scenario. Total exposure to dioxins and dioxin-like substances in the 0.3 E% LC n-3 PUFAs scenario ranges from 2.31 pg TEQ/kg bw/day at the 5th percentile, over 4.37 pg TEQ/kgbw/day at the 50th percentile to 8.41 pg TEQ/kgbw/day at the 95th percentile. In the 0.46 E% LC n-3 PUFAs scenario, 5, 50 and 95th percentile are exposed to 2.74, 5.52 and 9.98 pg TEQ/kgbw/day, respectively. Therefore, if the recommended LC n-3 PUFAs intake would be based on fish consumption as the only extra source, the majority of the study population would exceed the proposed health based guidance values for dioxins and dioxin-like substances.